Technical Document #29: Software Engineering - Comprehensive Overview

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Code review examines code changes before merging. It improves code quality, catches bugs, and shares knowledge across the team.

Technical debt represents the cost of choosing easy solutions over better ones. It accumulates over time and slows development velocity.

Continuous integration (CI) automatically builds and tests code changes. It catches integration issues early in the development process.

Refactoring improves code structure without changing behavior. It makes code easier to understand, maintain, and extend.

Continuous deployment (CD) automatically deploys code changes to production. It enables rapid, reliable releases.

API design defines how software components communicate. RESTful APIs and GraphQL are popular API design approaches.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Key Takeaways:

- Technical debt represents the cost of choosing easy solutions over better ones.
- Continuous integration (CI) automatically builds and tests code changes.
- Microservices architecture structures applications as independently deployable services.